University of Science and Technology

Zewail City

CSAI 205: Fundamentals of Circuits and Electronics

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Project 1: Part 2

Multi-Output Regulated Power Supply (220 V AC to 12 V, 9 V and 5 V DC)

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1. Objective

The goal of this part is to design, simulate and build a regulated DC power supply that converts the 220 V AC mains voltage into three stable DC outputs: +12 V, +9 V and +5 V. The supply is made of a step-down transformer, a bridge rectifier, a smoothing capacitor, three linear voltage regulators and an LED indicator on each output.

The circuit was first simulated in Proteus and then built and tested on a breadboard.

2. Components Used

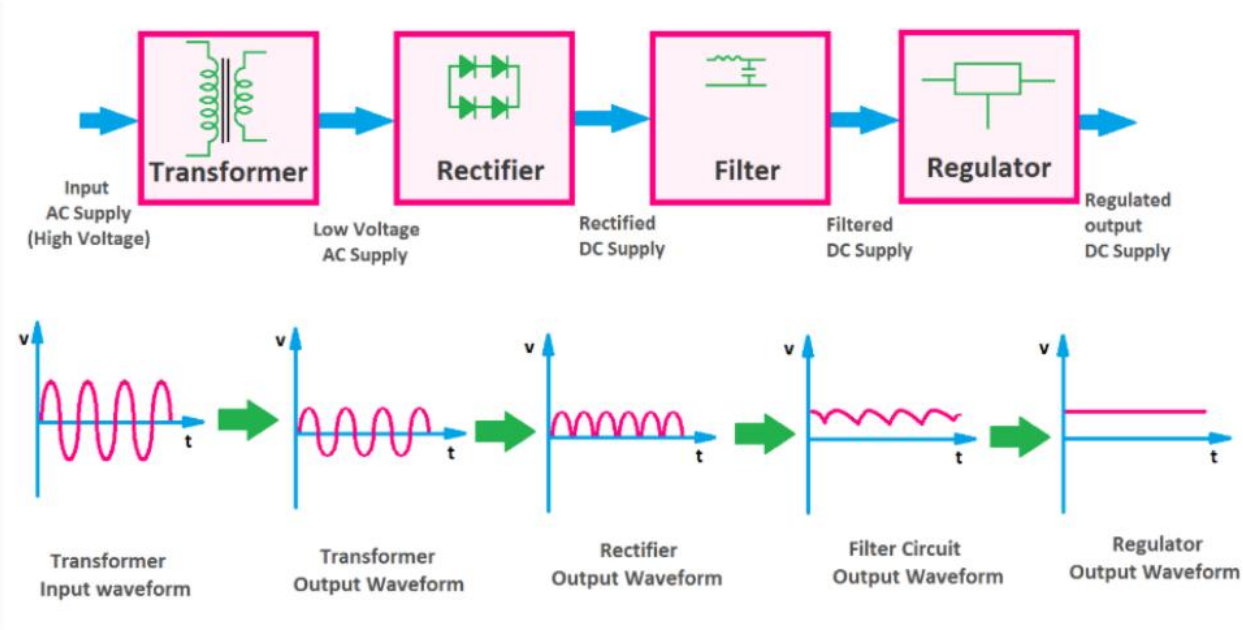
Component	Quantity used
Center-tapped step-down transformer 220 V to 6-0-6 V (500 mA marked on our unit)	1
AC power cord 220 V	1
Silicon diodes 1N4001 (bridge rectifier)	4
Electrolytic capacitor 1000 µF / 50 V	1
Capacitors 1 nF	3
Voltage regulators 7812, 7809, 7805	1 of each
Resistors 430 Ω, 330 Ω, 220 Ω	1 of each
5 mm LEDs	3
Breadboard (830 tie points) and jumper wires	1 / 10+

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3. Background and Block Diagram

The supply works in a chain of stages, and each stage solves one problem:

220 V AC → Transformer → Bridge rectifier → Filter capacitor → Regulators → 12 V / 9 V / 5 V



3.1 Transformer

The transformer steps the mains voltage down to a safe AC level (about 12 V RMS between the two outer secondary terminals) and electrically isolates the low-voltage side from the mains. Its peak value is about 12 × 1.414 ≈ 17 V. We used the two outer secondary terminals; the center tap (0) was left unconnected.

3.2 Bridge rectifier (D1 to D4)

Four diodes convert AC into pulsating DC. When terminal A is positive, current flows through D1, the load and D4. When terminal B is positive, current flows through D2, the load and D3. In both cases the current passes through the load in the same direction. Two diodes conduct at any time, so about 1.4 V is lost, giving a peak of roughly 17 − 1.4 ≈ 15.6 V at a pulsating frequency of 100 Hz.

3.3 Filter capacitor (C1 = 1000 µF)

The capacitor charges to the peak of each pulse and supplies the load between pulses, turning the pulsating DC into an almost constant voltage. a load current of about 60 mA, f = 100 Hz and C = 1000 µF, the ripple is about 0.6 V. The capacitor is polarized and must be connected with the correct polarity.

3.4 Voltage regulators (7812, 7809, 7805)

Each regulator takes the filtered voltage and produces a fixed output (12 V, 9 V or 5 V) even if the input changes slightly. The input must be about 2 V higher than the output (dropout), so the 7812 needs roughly 14 V at its input. The extra voltage is dissipated as heat: P = (Vin − Vout) × I. With LED loads of about 20 mA the heat is small, so no heat sink was needed.

3.5 Small capacitors (C2 to C4 = 1 nF)

Placed at each regulator output, they filter high-frequency noise and help keep the regulator stable.

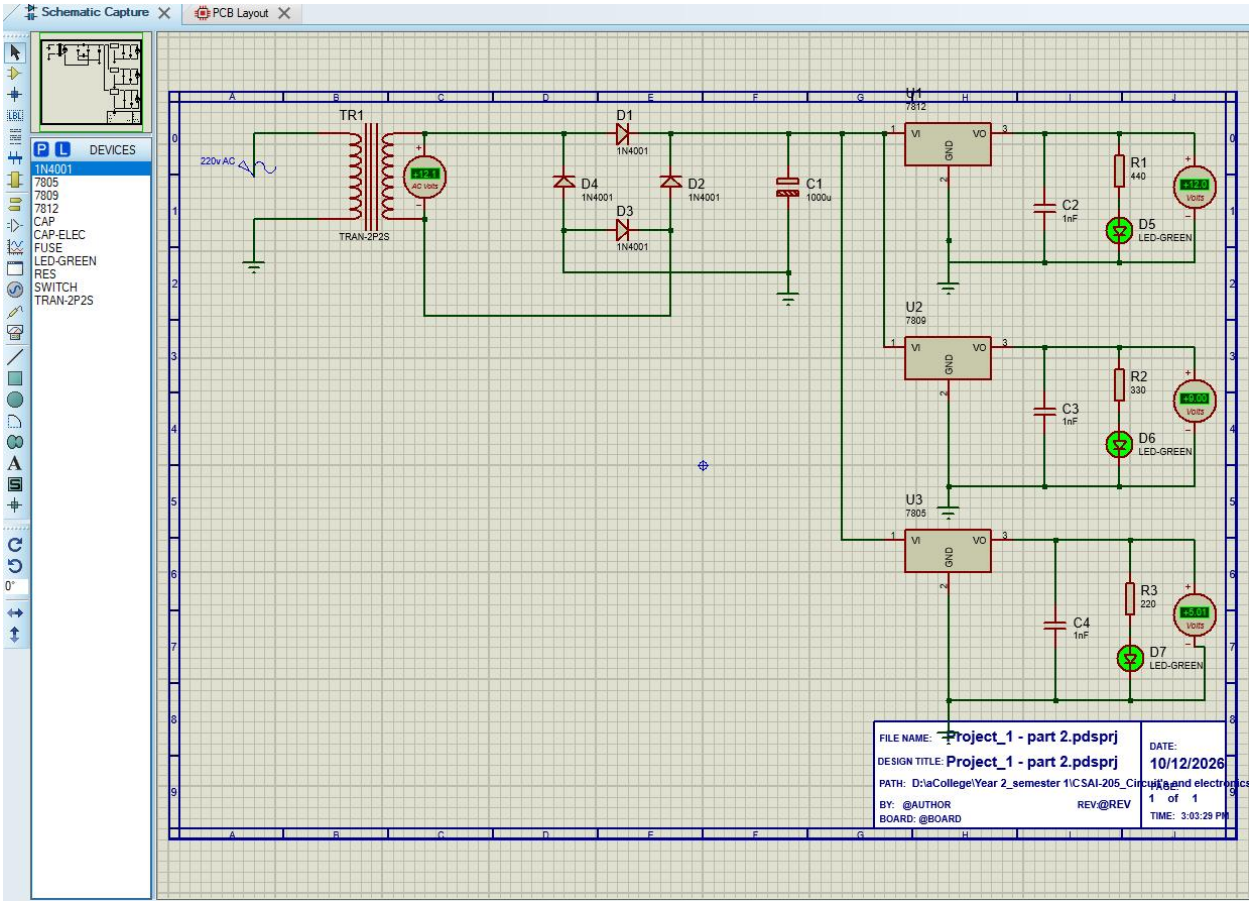
3.6 LED indicators

Each output has a resistor and LED to show that the regulator is working. The resistor sets the LED current: I = (Vout − V_LED) / R, with V_LED ≈ 2 V.

Output	Resistor	LED current (calculated)
12 V	430 Ω	(12 − 2) / 430 ≈ 23 mA
9 V	330 Ω	(9 − 2) / 330 ≈ 21 mA
5 V	220 Ω	(5 − 2) / 220 ≈ 14 mA

All grounds in the circuit are connected together to form a common ground reference.

4. Proteus Simulation



Simulation results:

Measurement point	Expected
Transformer secondary (AC)	≈ 12 V
Across C1 (DC)	≈ 15 to 16 V
7812 output	12 V
7809 output	9 V
7805 output	5 V

5. Hardware Implementation

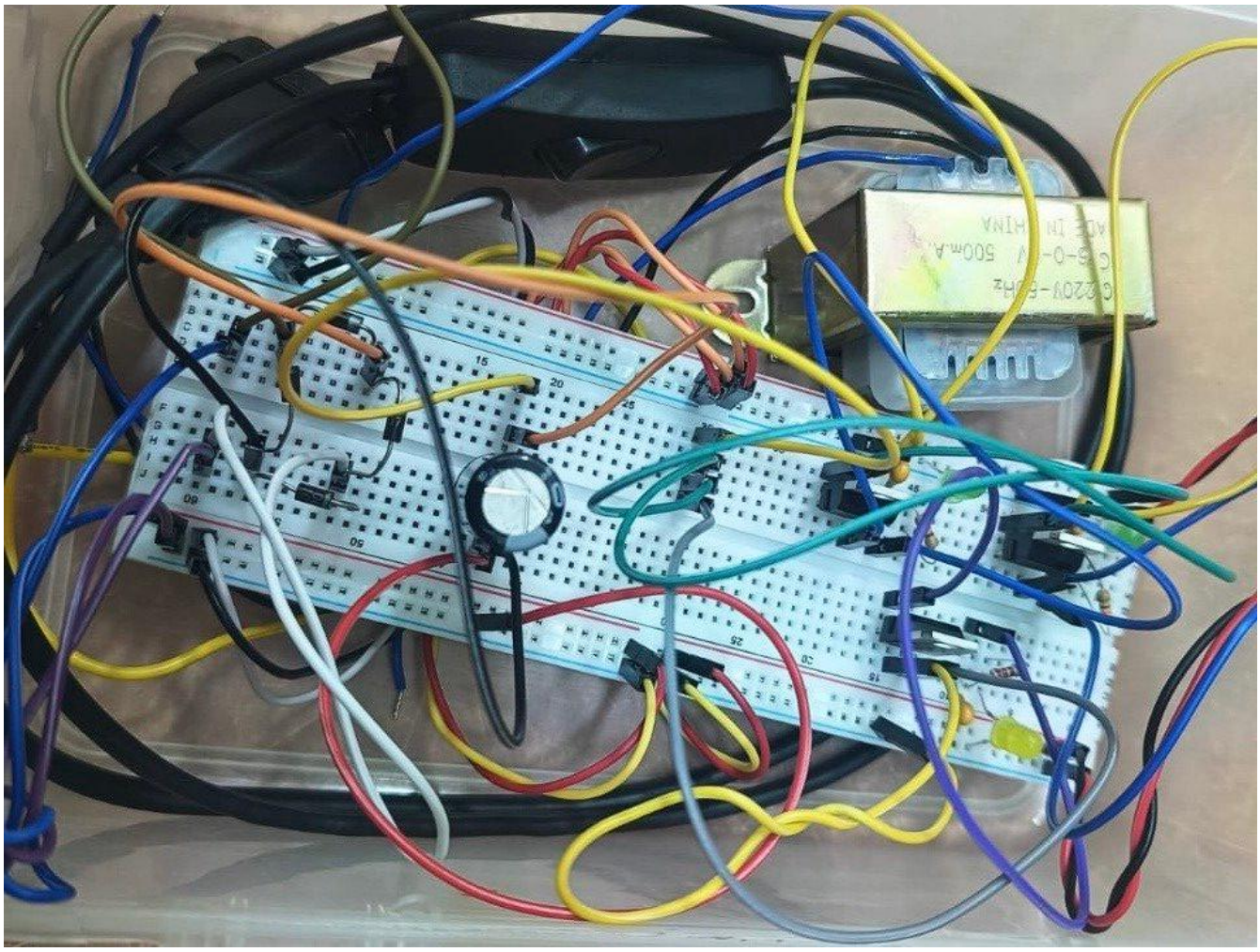
5.1 Identifying the transformer wires

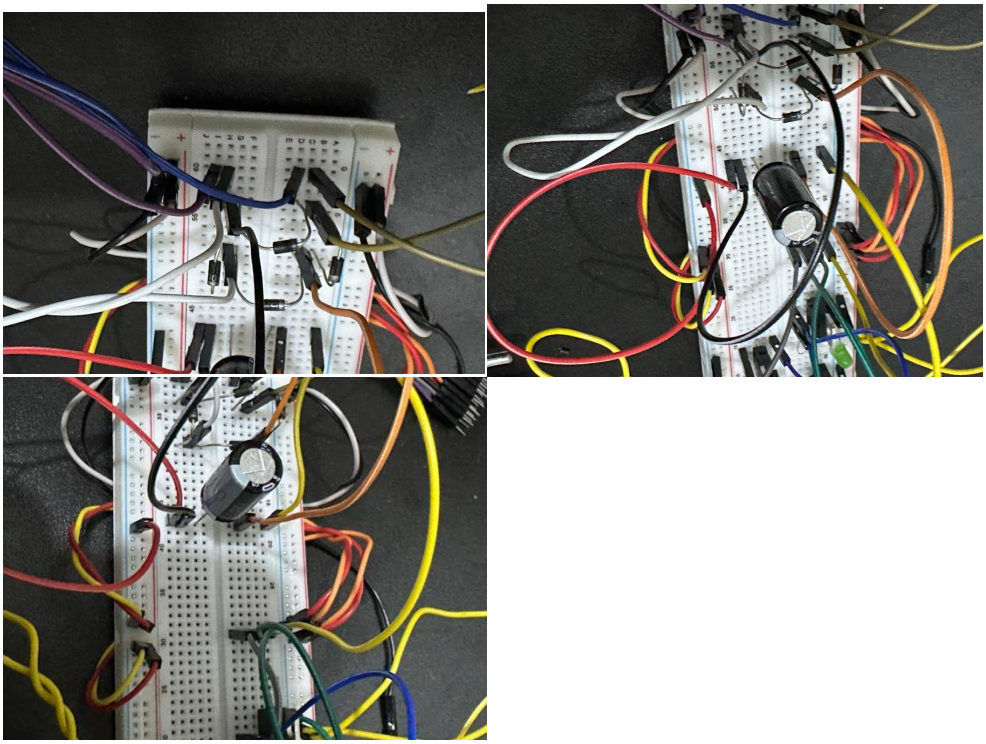
The transformer has two yellow wires, two blue wires and one black wire. Before connecting anything, we identified them with a multimeter in resistance mode:

- Between the two yellow wires we measured about 400 Ω (high resistance), so they are the primary winding, which connects to the 220 V mains.
- Between the blue wires and the black wire the resistance was small, so they belong to the secondary winding. The two blue wires are the outer terminals (6 V and 6 V, about 12 V AC between them) and the black wire is the center tap.

5.2 Building the circuit on the breadboard

1. The two blue wires were connected to the two AC inputs of the bridge (points A and B). The black center-tap wire was left unconnected and insulated.
2. The four 1N4001 diodes were placed as a bridge: D1 and D2 with their cathodes (striped end) on the positive rail, D3 and D4 with their anodes on the ground rail.
3. The 1000 µF capacitor was placed between the positive rail and ground, with the negative marking toward ground.
4. The three regulators were placed in separate columns of the breadboard. Looking at the printed face with the legs pointing down, the legs from left to right are input, ground and output. Inputs go to the positive rail and grounds to the ground rail.
5. A 1 nF capacitor was placed between each regulator output and ground.
6. At each output, a resistor (430 Ω, 330 Ω or 220 Ω) in series with an LED was connected to ground. The long LED leg (anode) is toward the resistor.





6. Problems Faced and Solutions

6.1 Connecting the two primary wires to the AC power cord

Problem. The two primary (yellow) wires of the transformer are thin, bare-ended leads. They have no connector, and they carry the 220 V mains, so they cannot be pushed into the breadboard or simply twisted loosely onto the power cord. A loose or exposed connection at this point is both unreliable and dangerous.

6.2 Identifying the transformer wires

Problem. The wires are not labeled, so it was not clear which pair was the primary and which was the secondary.

Solution. We measured the resistance between the wires with a multimeter. The yellow pair read about 400 Ω (primary) and the blue and black wires read much lower (secondary), which confirmed the color coding before anything was connected.

6.3 Identifying the regulator pins

Problem. The regulators have three identical-looking legs, and reversing the input and output can damage the part.

Solution. We checked the pin order for the 78xx series: with the printed face toward us and the legs pointing down, the pins are input, ground and output from left to right.

7. Safety Precautions

- The 220 V mains is dangerous. It was present only on the transformer primary; the breadboard worked entirely at low voltage.
- All mains connections were insulated, and the circuit was disconnected from power before any change was made.
- The connection to the mains was checked by the lab supervisor before power was applied.
- The electrolytic capacitor polarity was checked before power-on, because a reversed capacitor can overheat or burst.
- The unused center-tap wire was insulated so it could not touch any other conductor.

8. Conclusion

We built a multi-output regulated power supply that converts 220 V AC into 12 V, 9 V and 5 V DC, using a transformer, a diode bridge, a smoothing capacitor and three linear regulators. We understood the role of each stage and how the voltage changes from one stage to the next. The project also showed the importance of identifying components correctly (transformer wires and regulator pins) and of working safely with mains voltage.